SAD Short Note

(S1)

Information system

- An arrangement of Computer and Telecommunication Technology to support and improve day to day operations

Main components of an Information System

- Input (collects data)
- Processing (process data)
- Output (disseminates data)
- Feedback

Computer Based Information Systems(CBIS)

- hardware
- software
- databases
- telecommunications
- people
- procedures

System Development Environment

Stakeholders

- Any person who has an interest in an information system and its outputs.

Mainly **five types**

- **Systems User**
- **Systems Owner**
- **Systems Analyst**
- **Systems Designer**
- **Systems Builder**

Systems User

A "**customer**" who will use an information system by,

- capturing
- validating
- entering
- responding to
- storing
- exchanging

Systems Owner

Person or organization having responsibility for the

- Developing
- Operating
- Maintaining the information system.

System builders

Are technical specialists involved with

- Constructing
- Testing
- Delivering the system into operation

Systems Designer

- specializes in creating systems in adherence to the client or company's needs.

They design the system including

- Databases
- Inputs
- Outputs
- Screens
- Network
- Software to meet the users requirements

System analysts

- are people who understand both business and computing

Role of system analyst

- Identify the problem
- Analyze and understand the problem
- Identify the solution requirements
- Identify alternative solutions
- Design and implement the best solution
- Evaluate the result
- defining user requirements.
- gather the necessary data
- develop models for new systems.

The System analyst's skills

Interpersonal skill:

- **This skills deal with relationships and the interface of the analyst with people in business.**

Communication:

- **Communication is not just reports, telephone conversations, and interviews.**

Understanding:

- **Identifying problems and assessing their effects**

Teaching:

- **Educating people in use of computer system**

Selling:

- **Selling ideas and promoting innovations in problem solving using computers.**

Technical skills:

- **Focus on procedures and techniques for operations analysis, systems analysis ,and computer science.**

Creativity

- **Helping users model ideas into concrete plans**

Problem solving

- **Reducing problems to their basic levels for analysis**

Knowledge computer & the business functions

Technical Skills cont..

- **Project management:**
  Scheduling, performing well
- **Dynamic interface:**
  Blending technical and nontechnical considerations
- **Questioning attitude and inquiring mind:**
  Knowing how a system works.

Analyst's Personal Attributes

- **Authority:**
  The confidence to "tell" people what to do
- **Communication skills:**
  Ability to focus on a problem area for logical solution.
- **Creativity:**
  Trying one's own ideas for developing
- **Responsibility:**
  Making decisions on one's own
- **Varied skills:**
  Doing different projects and handling change.

analyst's personal skills

- **Choosing a team to conduct a project**
- **Leading a team**
- **Giving technical ideas**
- **Obtaining information**

Types of Information Systems

Classification by

mode of processing

Batch processing systems:
Transactions are collected as they occur

On-line batch systems:

- *Transaction information is captured by on-line*

On-line Real-time systems

- *Capturing transaction data as well as updating & generating reports*

System Objectives

Type of information System

Transaction Processing System (TPS)

Management Information System (MIS)

- MIS is an application that provides management-oriented reporting.

Decision Support System (DSS)

- Provides decision-oriented information to the user.
- Interactive computer-based modeling process

Executive Information System (EIS)

- indicate issues of importance to the organization
- They integrate data from all over the organization (into graphical indicators and controls)

Expert Systems (ES)

- Simulates the "thinking " of the expert.
- programmed decision making information system
- indicate new directions the company may take

Communications and Collaboration Systems

Effective communications between,

- Workers
- Partners
- Customers
- Suppliers

Office Automation System

It supports the **wide range** of business office activities.

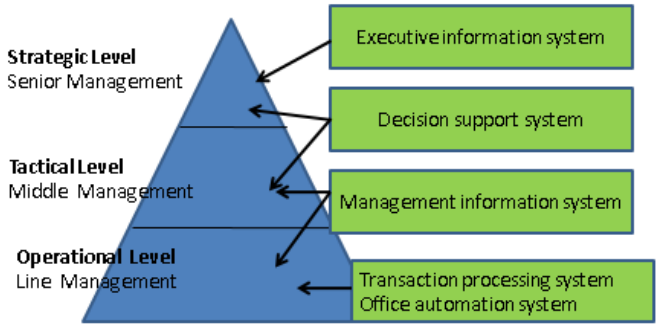
- Work group computing
- Work group scheduling
- E-mail
- Electronic document

Geographic Information System (GIS)

A system designed to

- capture,
- store,
- manipulate,
- analyze,
- manage,
- present (all types of spatial or geographical data)

information systems by management level.



Legacy Systems

- potentially problematic
- often run on obsolete hardware
- hard to maintain, improve and expand

<u>Upgrading a Legacy System</u>

- Force to change
- Many complex legacy systems are yet to
- be upgraded to new technologies because of
  - Cost
  - Skills
  - People required

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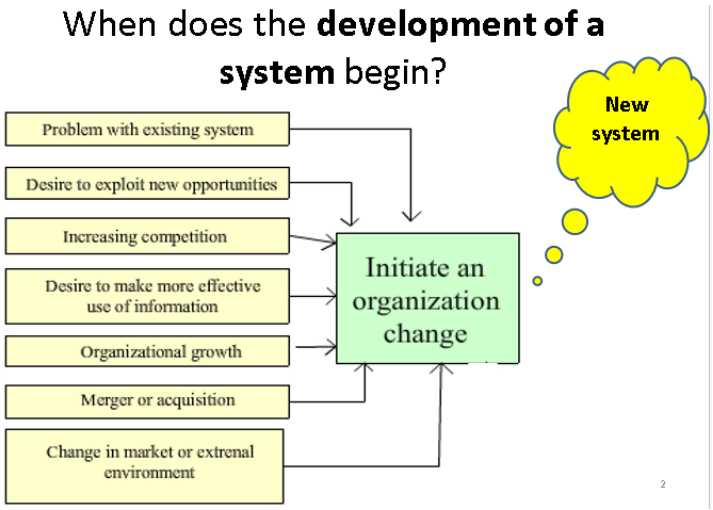
(S2) **Systems Development Life Cycle (SDLC)**



It is a logical process by build information systems.
Consists of several phases used to

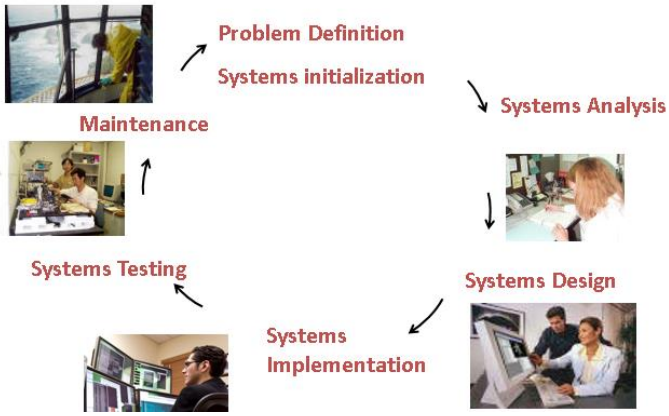
- plan,
- execute
- control

the system development projects.

We **need** a life cycle in systems development to,

- **ease the process**
- **build high quality**
- **work effectively and efficiently**
- **avoid failures**
- **maintain and enhance cost effectively**

System Development Life Cycle (SDLC)



- **Main phases of SDLC:**
  - **Systems Investigation*(Problem Definition*)**
  - ***Systems Analysis***
  - ***Systems Design***
  - ***Systems Implementation***
  - ***Systems Testing***
  - ***Systems Maintenance*** **pg6**

1.Systems Investigation(Problem Definition)

This phase identifies and defines a need for the new system. Called project's Terms of Reference (TOR).

Terms of Reference (TOR)

Project goals

The system provides what the users expect

Project bound

The project defines what parts should be changed and what should remain the same.

Project limits

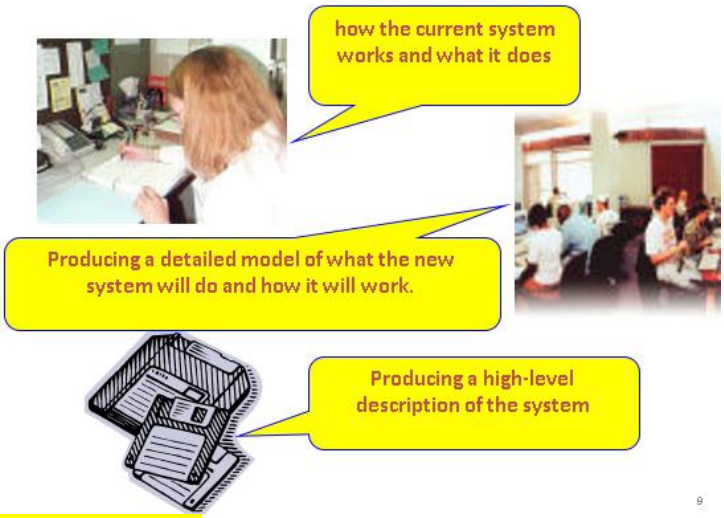
Resources to be provided for the project (resource constraints).

2. Requirement Analysis (System Analysis)

business requirements, expectations, and priorities for a solution to the business problem

phase produces

- detailed analysis model
- requirements model
- system requirements



3. Systems Design

Produces a design specification for the new system.

Things done during the design phase:

- Identify suitable hardware
- Specify new programs or changes to existing programs
- Specify new database or changes to existing database

4. System Implementation

systems implementation phase,

- **individual system components**
- **user interfaces**
- **a database**
- **data and tools**

5. System Testing

- Individual program modules are tested by their developers.
- Integration testing is performed to check whether the modules can be integrated.
- Designing test cases that test all conditions of the system input.

6. Maintenance

In this phase, the following problems are corrected.

- Fix any bugs and problems found by users
- Eliminate errors in the system during its lifetime.
- Tune the system into any variation in its working environment.
- Information system planners must always plan for resource
- availability to carry out these maintenance functions.

(S3)

Major Components of Systems Development

- Methodology
- Modeling Methods or Techniques
- Tools

Methodology

- A very **formal** and **accurate** system development process that defines a set of
  - Activities
  - Methods
  - Best practices
  - Deliverables
  - Automated tools

- Provides the framework
- Has a predefined set of steps

Eg:

- SSADM (structured methodologies)
- YSM (Yourdon Systems Method)
- STRADIS (Structured Analysis, Design, and Implementation of Information Systems) **page 8**

Methodology

Modeling Methods Or Techniques

Eg: Class diagram, usecase diagram

Tools

Eg: Rational rose

Modeling method

- A set of techniques used to implement a Methodology.

Data Flow Diagrams

- A process model
- The flow of data through a system and the work performed are represented.

Entity Relationship Diagrams

- A data model
- Depict data in terms of entities and relationships described by the data
- Consists of several notations

Structure Charts etc.

Tools

- Tools are Software systems
- it assists analysts and designers to build information systems.

General Aim :

- Increase the quality of software
- Tools will support methodologies
- Decrease the human effort to develop the software

e.g. Easy Case, Rational Rose

A software process

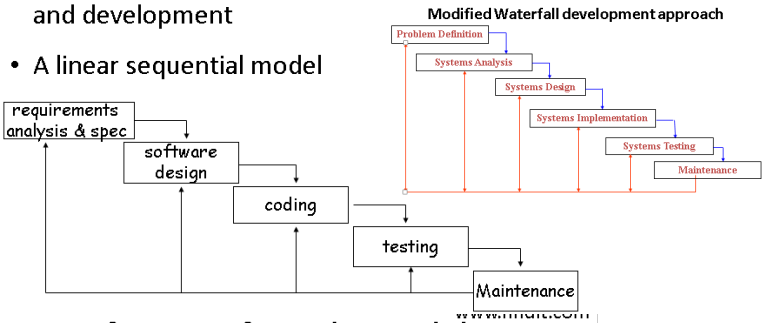
- is the set of activities and associated outcomes that produce a software product.

Software process models

- Waterfall model
- Prototyping models
- Evolutionary models
- The spiral model
- Formal development
- Incremental development
- Rapid Application Development
- Unified Process
- Agile Process
- Extreme Programming (XP)

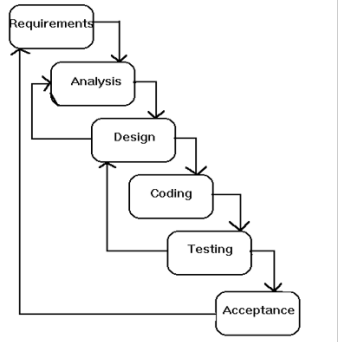
The Waterfall model

- Separate and distinct phases of specification and development
- A linear sequential model



Software Life Cycle Models

- Waterfall lifecycle



The waterfall model (Systems Development Life Cycle)

Waterfall Strengths

- **Easy to understand, easy to use**
- **Provides structure to inexperienced staff**
- **Milestones are well understood**
- **Sets requirements stability**

There are **several problems** with **Waterfall** approach.

- It has a rigid design
- Inflexible
- It has a top-down procedure
- No phase can be repeated
- Time consuming

Modified Waterfall Development Approach

- Uses **the same phases**
- Allow some of the stages to **overlap**
- Progress is more **difficult** to track.

When to use the Waterfall Model

- Attractive for large problems
- Projects which have clear and stable requirements.

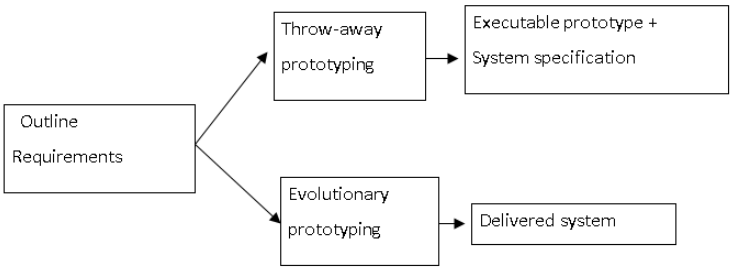
Prototyping

- A prototype can be used to clear the vague requirements.
- should be evaluated with user participation.

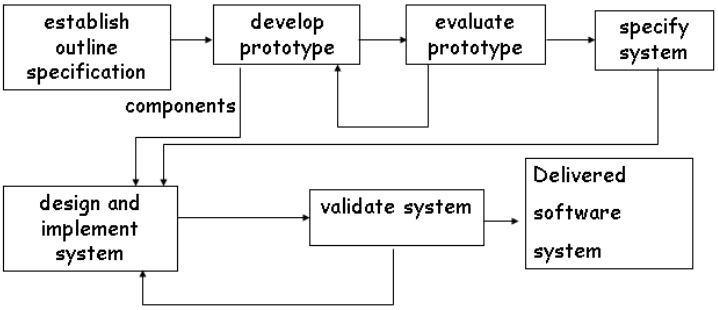
two types of Prototyping techniques

- Throw-away Prototyping
- Evolutionary Prototyping

Throw-away and Evolutionary Prototyping



Throw-away Prototyping



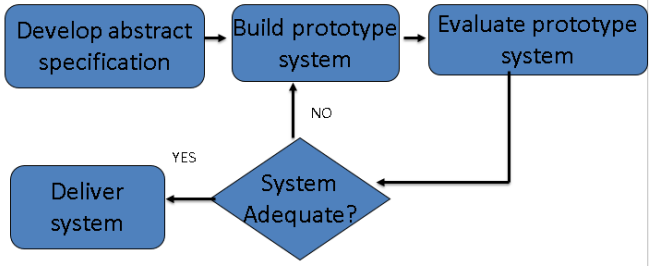
Throw-away Prototyping

- A throw-away prototype is a cheap, fast prototype that is designed to model an idea or feature

Problems with Throw-away Prototyping

- Developers may be pressurised by the users to deliver it as a final system!
- All the man-hours of putting together the throw-away prototypes are lost, unlike the evolutionary approach.

Evolutionary Prototyping



Evolutionary Prototyping Strengths

- Developers learn from customers
- A more accurate end product
- Unexpected requirements accommodated
- Allows for flexible design and development

Evolutionary Prototyping Disadvantages

- Can be costly to use.
- Management complexity is more.
- Not suitable for smaller projects.
- Highly skilled resources are required for risk analysis.

Prototyping Weaknesses

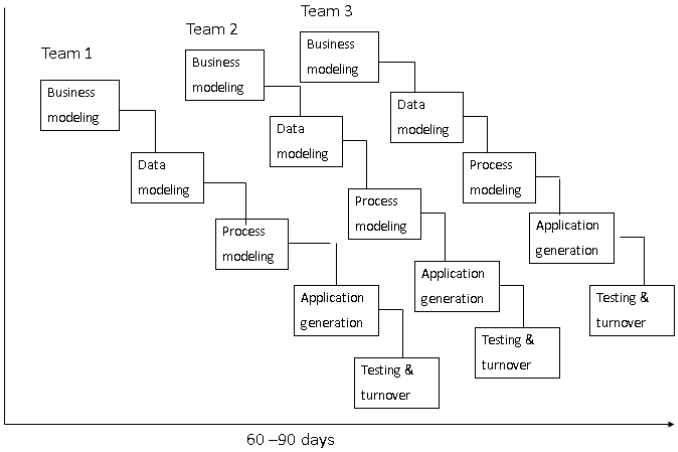
- **Prototyping can lead to false expectations.**
- **Prototyping can lead to poorly designed systems.**
- **Overall maintainability may be overlooked**
- **The customer may want the prototype delivered.**
- **Process may continue forever (scope creep)**

The RAD model

- (RAD) is an incremental software development process model that emphasizes an extremely short development cycle.

The RAD Model

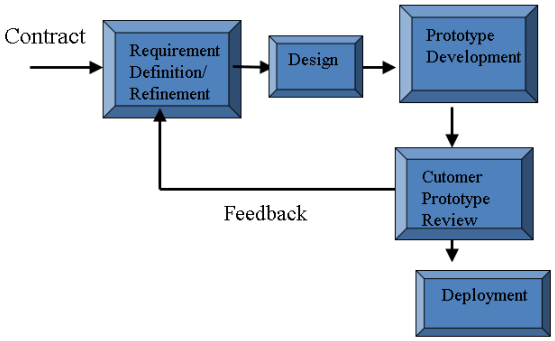
www.hndit.com



Processes in the RAD model

- **Business modelling**
- **Data Modelling**
- **Process Modelling**
- **Application generation**
- **Testing and turnover**

Cont. RAD lifecycle



Problems with the RAD model

- Requires highly skilled developers/designers.
- High dependency on modeling skills
- The only system that can be modularized can be built using RAD.

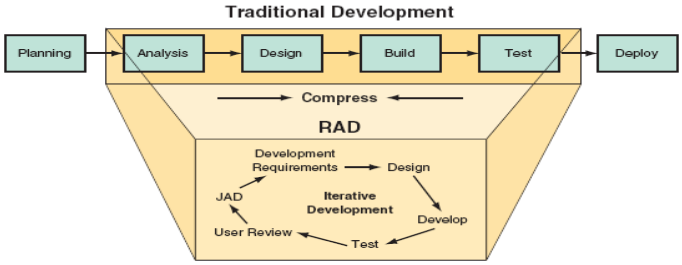
Depends on strong team and individual performances for identifying business requirements.

RAD lifecycle

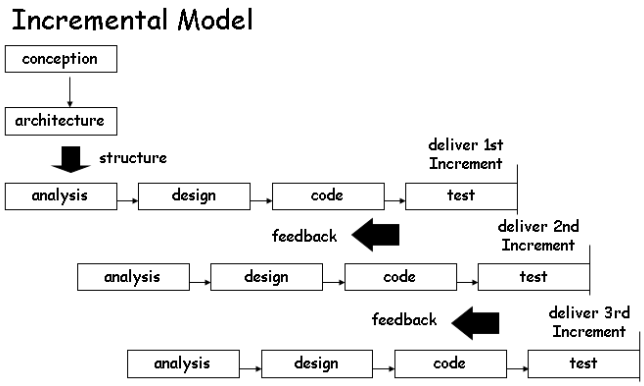
- User requirements are refined.
- A system is designed.
- A prototype is developed.
- The prototype is reviewed by end-user.
- User provide feedback.
- The process is repeated.

Advantages of RAD

- Can speed up systems development.
- Users thoroughly involved from the start.
- Improves the process of rewriting legacy applications.



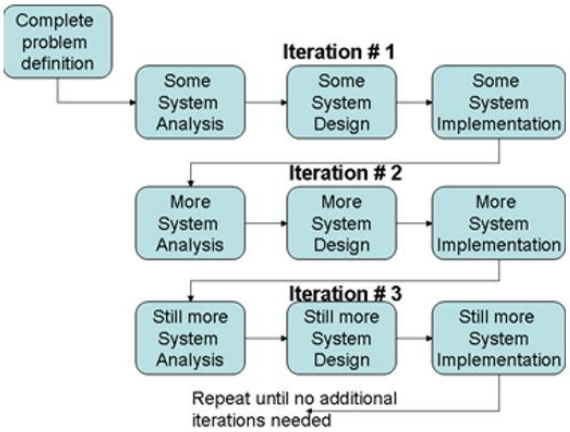
Agile methodology



Incremental Development

- Plans and documentation must be produced
- The most important part of the system is first delivered
- more manageable than evolutionary prototyping
- model involves developing the system in an incremental fashion.

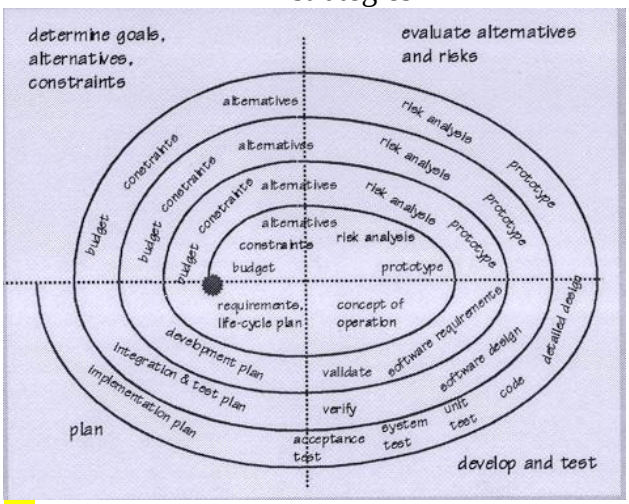
Iterative Development Approach



- Completes the entire IS in successive iterations.
- Each iteration does some
  - *analysis*
  - *design*
  - *construction*
- Allows versions of usable information to be delivered in regular and shorter time frames.
- The iteration process helps to develop a part of the new system

Spiral model

- quadrant 1
  - alternatives and constraints.
- quadrant 2
  - analyzed form the viewpoint of risk
- quadrant 3
  - where the traditional waterfall model phases are put into practice.
- Quadrant4
  - the results of the risk-reduction strategies



Spiral Quadrant

Determine

- **Objectives: functionality, performance,**
- **Alternatives: build, reuse, buy**
- **Constraints: cost, schedule, interface**
- **Resolve risks (evaluate if money)**
- **Study alternatives (relative to objectives and constraints)**
- **Identify risks (lack of experience, new technology)**

Develop next-level product

- **Typical activates:**
  - **Create a design**
  - **Review design**
  - **Develop code**
  - **Inspect code**
  - **Test product**

Plan next phase

- **Typical activities**
  - **Develop project plan**
  - **Develop configuration management plan**
  - **Develop a test plan**
  - **Develop an installation plan**

Spiral Model Strengths

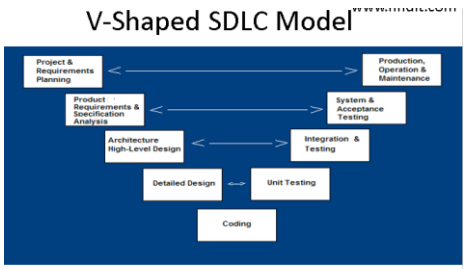
- **Critical high-risk functions are developed first**
- **The design does not have to be perfect**
- **Users can be closely tied to all lifecycle steps**
- **Early and frequent feedback from users**

Spiral Model Weaknesses

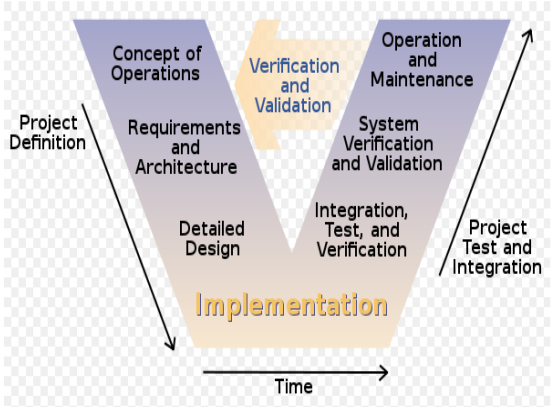
- **The model is complex**
- **Risk assessment expertise is required**
- **hard to define objective**
- **spent for evaluating risks**

When to use Spiral Model

- **Users are unsure of their needs**
- **Requirements are complex**
- **New product line**
- **For medium to high-risk projects**



Testing of the product is planned in parallel with a corresponding phase of development

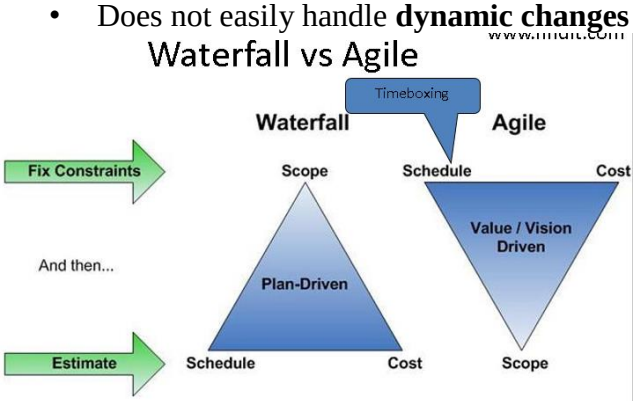


V-Shaped Strengths

- Each deliverable must be testable
- Easy to use
- Project management can track progress by milestones
- Emphasize planning for **verification and validation**

V-Shaped Weaknesses

- Does not easily handle **concurrent events**
- Does not handle **iterations** or phases
- Does not contain **risk analysis** activities
- Does not easily handle **dynamic changes**



Agile methodology

- is a way to manage a project by breaking it up into several phases.
- Phases: planning, requirements analysis, design, coding, testing, & documentation

Agile Vs. Iterative

- Time period is measured in weeks rather than months
- Work is performed in a highly collaborative manner
- Treating time period as a strict timebox

Agile Vs. Waterfall

- Produce completely developed and tested features within a phase
- Emphasis is on obtaining the smallest workable piece of functionality
- Continually improving/adding further functionality

(S6) Problem Identification Methods

Problem Definition

- **Identifying Requirements**
  first, clearly define what the system must do.
  Requirements may be functional or non-functional

Requirements Elicitation

- *process of seeking, uncovering, acquiring, and elaborating requirements for computer-based systems.*

System Requirements

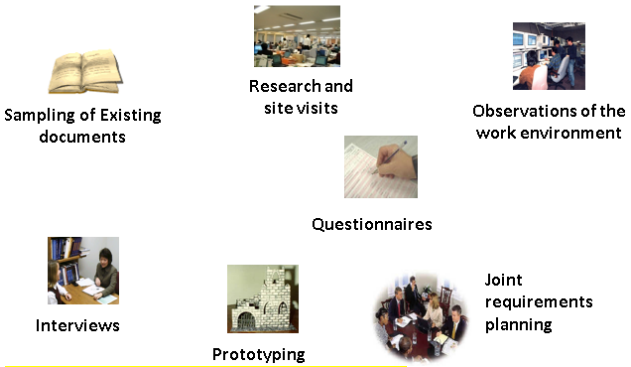
Specify what the information system must do, or what property / quality the system must have

System Requirements

Functional Requirements	Non functional Requirements
Specify what the information system must do	Specify a property / quality the system must have

Example : Library system
Functional- *borrowing books, returning process*
Non functional- *security level, interfaces*

Fact Finding techniques



Requirement Discovery Methods (Fact Finding Techniques)

- It is the formal process of using techniques to collect information about systems requirements

Methods

Sampling of existing documents

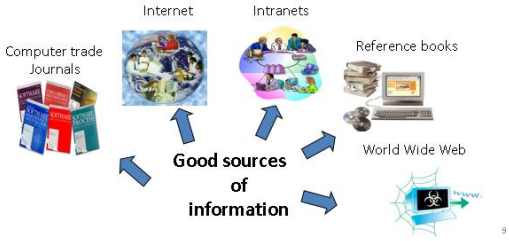
You can get good idea by studying existing

- *Documents*
- *Forms*
- *files*

Research and site visits

2. Research and site visits

- Thoroughly, research the problem domain.
- Identify the material that are relevant and reliable



Observations of the work environment

- ***System Analyst participates in or watches a person perform activities to learn about the system.***
- **Advantages**
  - **Highly reliable.**
  - **Relatively inexpensive**
- **Disadvantages**
  - **People usually feel uncomfortable when being watched**

Questionnaires

A questionnaire is a research tool featuring a series of questions used to collect useful information from respondents.

- **Advantages**
  - **Most questionnaires Can be answered quickly.**
  - **Allow individuals to maintain anonymity.**
  - **Relatively inexpensive way of gathering data.**
  - **Responses can be tabulated and analyzed quickly etc.**
- **Disadvantages**
  - **Number of responses is often low.**
  - **Difficult to prepare**
  - **No guarantee that an individual will answer or expand on all the questions**

Type of questionnaires

- **Free-format:**
  Answer in the space provided after the question.
- **Fixed-format:**
  required specific responses from individuals.
  1)Yes /No Questions
  2) Multiple choice questions
  3) Rating questions

Interviews

- *System Analyst collects information from individuals face to face.*
- ***Advantages***
  - *New ideas may arise*
  - *Ability to find the right candidate*
  - *Enables detailed assessment*
  - *Great source of information*
  - *Increase knowledge*
- ***Disadvantages***
- *Highly time-consuming*
- *Risk of personal bias*
- *Quick to judge*
- *Can easily form stereotypes*

- ***Types of Interviews***
  - *Unstructured interviews*
  - *Structured interviews*
- ***Types of interview questions***
  - *Open ended*
  - *Closed ended*

How to conduct an interview?

- **Select interviewees**
- **Prepare an interview guide**
  - **Use clear and concise language.**
  - **Avoid long and complex question**
  - **Avoid threatening questions**
  - **verify before you leave**
  - **Don't include your opinion as part of a question.**

Joint requirements planning

- *Highly structured group meeting are conducted to analyze problems and define requirements.*
- *JRP is a subset of a more comprehensive joint application development or JAD technique.*

Brainstorming

- *let the stakeholders come up with creative ideas or new approaches to a problem*

(S7) Feasibility Study

- *The measure of how beneficial an information system will be to an organization.*
- *Categories of feasibility tests,*
  - *Operational Feasibility*
  - *Cultural / Political Feasibility*
  - *Technical Feasibility*
  - *Schedule Feasibility*
  - *Economic Feasibility*
  - *Legal Feasibility*

Operational Feasibility

- It is a measure of **how well a solution meets the identified system requirements** to solve the problem.

Cultural (or Political) Feasibility

- deals with how the end users feel about the proposed system.

Technical Feasibility

- It is a measure of the **practicality** of a technical solution
- **availability** of technical recourses and expertise
- **three major issues**:
  - Is the proposed technology or solution practical?
  - Do we currently possess the necessary technology (Hardware/Personnel) ?
  - Do we possess the necessary technical expertise?

Schedule Feasibility

- A measure of how reasonable a **project time table** is.

Economic Feasibility

- It is a measure of **the cost-effectiveness of a project.**

Cost-benefit analysis

- **is always included both tangible and intangible items.**
- **Tangible items are those to which direct values can be attached**
  - *Equipment costs for the new system*
  - *Personnel cost*
  - *Material costs*
  - *Conversion cost*
  - *Other costs(consultant's cost)*
- **Intangible items, are those whose values cannot be exactly determined**

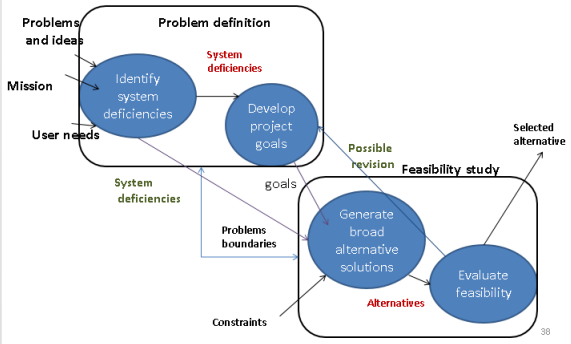
Determining whether a project is worthwhile

- ***Payback method***
- *defines the time required to recover the money spent on a project.*
- ***Present value method***
- *determine how much money it is worthwhile investing*

Legal Feasibility

- *Study to know if the proposed project conforms to the legal and ethical requirements.*
- *copyright law*
- *non-disclosure clauses*
- *code ownership*
- *labour laws*
- *foreign trade, and labour regulations*
- *Financial & Accounting standards*

Activities in problem definition and feasibility analysis



Feasibility Analysis of candidate systems

- ***Candidate System Matrix***

Candidate System Matrix www.hndit.cc

1) **Identify how the system will interact with people, and other systems**
2) **How data stores be implemented, inputs will be captured, outputs will be generate**
3) **How computer processes will be implemented**
4) **Identify how processed and data will be distributed**

	Canditate1	Canditate1	Canditate1
Stakeholders			
Knowledge			
Processes			
Communication			

- ***Feasibility Analysis Matrix***

Feasibility Analysis Matrix

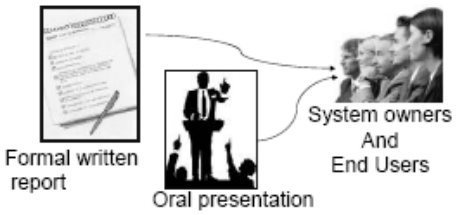
	Weighting	Candidate 1	Candidate 2	Candidate 3
Description				
Operational Feasibility				
Cultural Feasibility				
Technical Feasibility				
Economic Feasibility				
Legal Feasibility				
Weighted Score				

(S8)The System Proposal

A report/ presentation of a recommended solution
Intended for system owners and users..

The System Proposal

- A report / presentation of a recommended solution
- Usually a formal written report or oral presentation
- Intended for system owners and users



Written Report

- The most common method used by analysts
- Consists of
  - Primary Elements
    - Actual information
    - E.g. introduction, conclusion
  - Secondary Elements
    - Package the report in a way where the reader can easily identify the report and its primary elements
    - Add a professional polish.

Elements for a written report

Letter of transmittal
Title page
Tables of contents
List of Figures, illustrations, and tables
Abstract or executive summary
(primary elements are presented in this portion of the report)
Appendixes

Formal Presentation

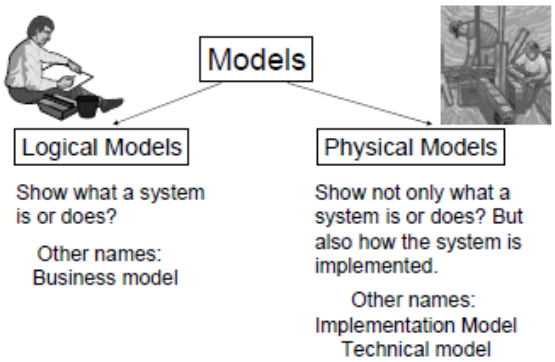
- A special meeting used to
  - sell new ideas
  - gain approval for new systems
  - verify conclusion
  - clarify facts
  - report the progress

- During this Feasibility Study phase
- Identifies candidate system solutions
- Analyses the solution for feasibility
- present the system proposal
- Report and oral presentation

(S9) System Design

Model

- Model is a presentation of reality.



Modelling Methods

- ***Data Flow Diagrams***
- A process model
- Depict the flow of data in system
- ***Entity Relationship Diagrams***
  – A data model
  – Consists of several notations
- ***Structure Charts etc.***

Process Modeling

- graphical representation of business processes or workflows.
- **Systems Analyst may first draw**
  Data Flow Diagram OR
  Documents Flow Diagram

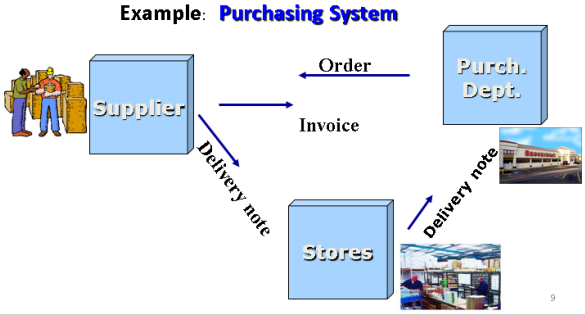
Logical and physical data flow design

- ***Logical DFDs-***
  that illustrate what occurs **without showing** how it occurs are known as logical DFDs.
- ***Physical DFDs-***
  DFDs that **show** how things happen or the physical components are called physical DFDs

Document Flow Diagrams

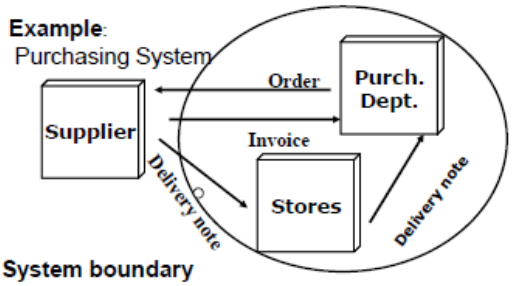
Used to examine the flow of documents within the existing system.

Example: **Purchasing System**



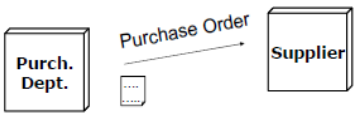
Document flow diagram

Example:
Purchasing System



System boundary

– snows
- where the document comes from,
- where it goes to , and
- what it is called.



Data Flow Diagram (DFD)

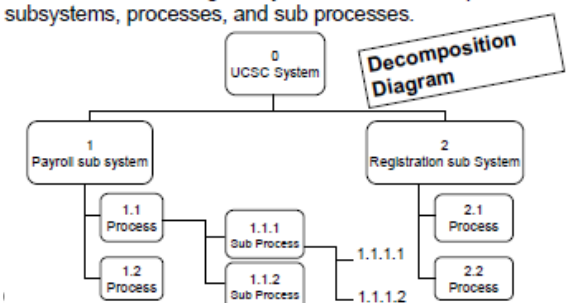
- Process Modeling Technique.
- Shows flow of data through a system and the processing performed by the system.
- Some analysts draw a **decomposition diagram** before DFD.

Decomposition

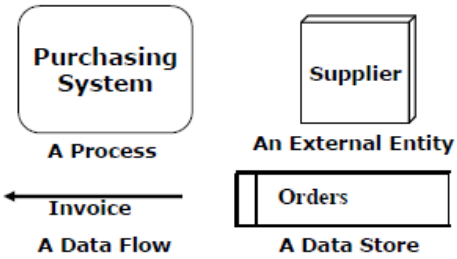
- Break a relation into multiple relations to bring it into an appropriate normal form.

Process Modeling…

Process Decomposition
Is a act of breaking a system into its component subsystems, processes, and sub processes.



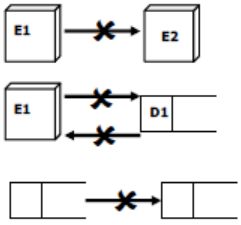
Elements in a DFD
(Gane and Sarson Symbols)



Data Flows

- **Data flow model the passage of data in the system and are represented by lines joining system components.**
- **Flows of data in the system can take place:**
  - **Between two processes**
  - **From a data store to a process**
  - **From a process to a data store**
  - **From entities to a process**
  - **From process to a entities**

Illegal Data Flows



Another Common error

Read Only ?

Context diagram

- **A common way to begin is to model, the whole system by one process.**
- **defines the scope of the system by identifying the system boundary**

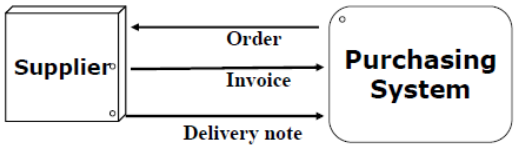
contains:

- **one process (***which represents the entire system***)**
- **all sources/sinks (***external entities***)**
- **data flows linking the process to the sources and sinks (***external entities***)**

Constructing a Context Diagram

- **identify and list sources/sinks (external entities)**
- **identify and list inputs to and outputs from sources/sinks (external entities)**
- **create context diagram**

Shows the top level function defined to cover the scope of the application.



Data Flow Diagrams

- Draw Context Diagram
- Level 0 (Top Level) Data Flow Diagram
- Level 1 Data Flow Diagram
- Continue up to elementary functions

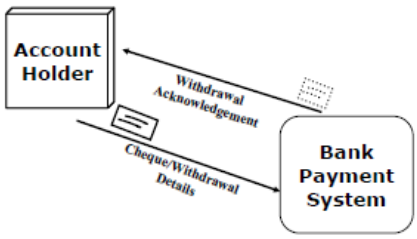
Bank Payment System

Consider a system in a bank whereby account holders get their withdrawals affected.

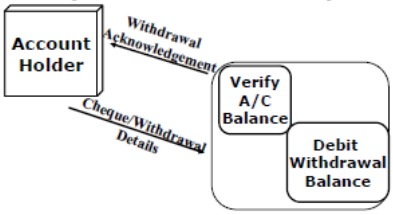
Whenever an account holder wants to withdraw some cash,
he presents a cheque or withdrawal slip.

The account is checked for appropriate balance.
If balance exists, the cash is paid and the account is updated.

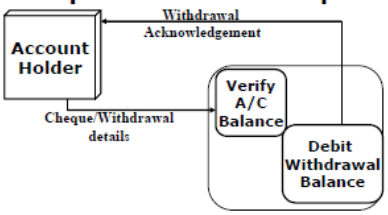
Context Diagram



Top Level DFD – Step1

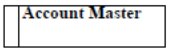


Top Level DFD – Step2



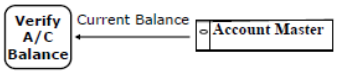
Top Level DFD – Step3

- Identify the Data Stores



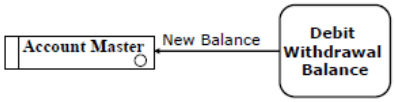
Top Level DFD – Step4

- Identify the other data flows.
  Get current balance

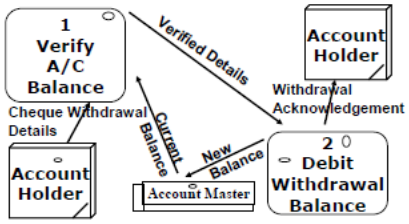


- Identify the other data flows.
  Transfer the verified details

- Identify the other data flows.
  update new balance



Top Level Diagram



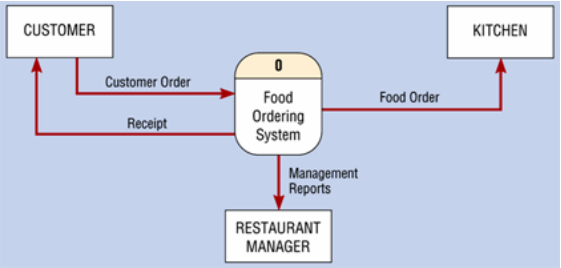
Level-0 Diagram

- **show one process for each major processing step or functional requirement**

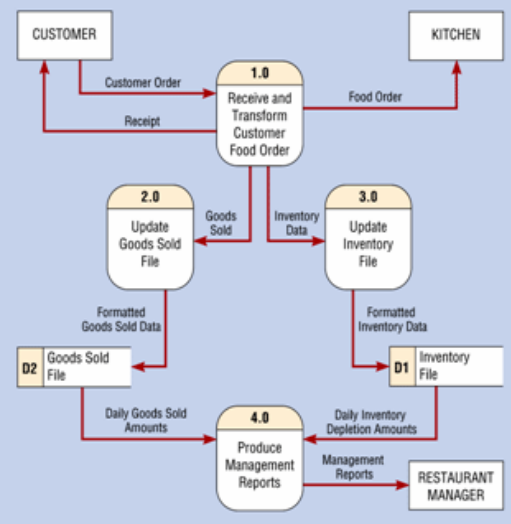
Drawing a Level-0 Diagram

- **list the major data stores**
- **list major business steps**
- **draw a segment for each business step**
- **assemble into single DFD**
- **re-organize until satisfied**
- **number processes**

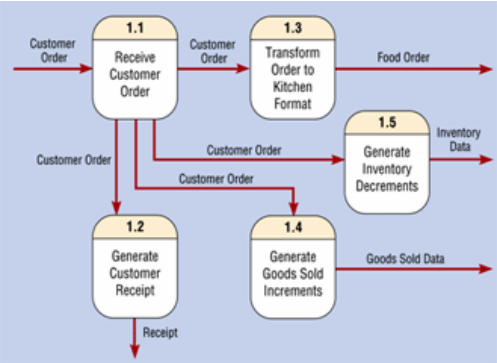
DFD context level diagram



DFD level 0 diagram



DFD level 1 diagram



Questions about Lower level diagrams

1. How deep? (how many levels?)

- *if the process has only one input or one output, probably cannot partition further;*

2. How broad? (how many processes on a level?)

- *7 ± two is a reasonable*
- *may temporarily place much of the system on a single diagram then re-draw into separate levels*

Quality Guidelines

- **Completeness**
  - **all components included & in project dictionary**
- **Consistency**
  - **between levels: balancing, leveling**
- **Timing considerations**
  - **assume system never starts and never stops**
- **Iterative nature**
  - **revisions are common**
- **Drawing primitives (lowest level)**
  - **when to stop?**

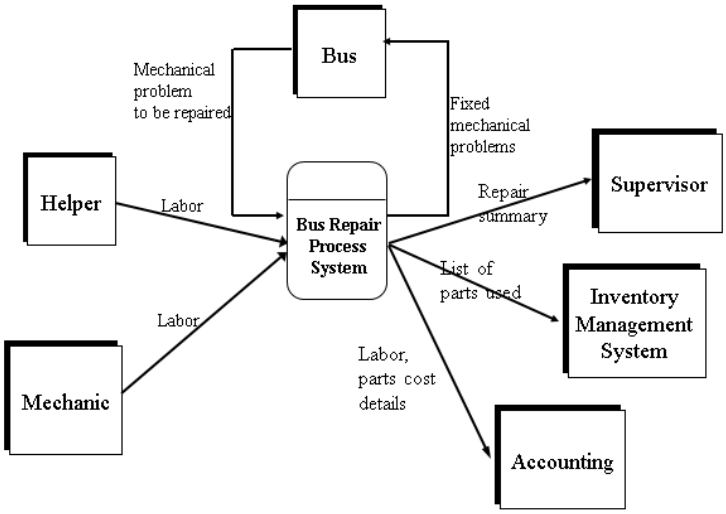
DFD Example: Bus Garage Repairs

- **Buses come to a garage for repairs.**
- **A mechanic and helper perform the repair, record the reason for the repair and record the total cost of all parts used on a Shop Repair Order.**
- **Information on labor, parts and repair outcome is used for billing by the Accounting Department, parts monitoring by the inventory management computer system and a performance review by the supervisor.**

- ***External Entities:***
- Bus, Mechanic,
- Helper,
- Supervisor,
- Inventory Management System,
- Accounting Department, etc.
- ***Key process*** ("the system"): performing repairs and storing information related to repairs
- ***Processes:***
  - Record Bus ID and reason for repair
  - Determine parts needed
  - Perform repair
  - Calculate parts extended and total cost
  - Record labor hours, cost
- ***Data stores***:
  - Personnel file
  - Repairs file
  - Bus master list
  - Parts list
- ***Data flows***:
  - Repair order
  - Bus record
  - Parts record
  - Employee timecard
  - Invoices

Bus Garage Context Diagram

www.hndi



What is a good data flow diagram?

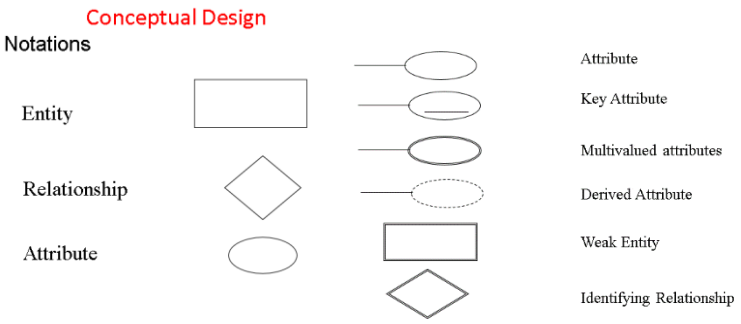
- **The absence of flowchart structures**
- **protection of data**
- **Good naming conventions**

Differences between flowcharts and data flow diagrams

- **DFDs are not program flow chats and should not include control elements.**
- **A good DFD should;**
  - *Have no data flows that split up into a number of other data flows.*
  - *Have no crossing lines*
  - *Not include flowchart loops of control elements*
  - *Not include data flows that act as signals to activate processes*

ER-DIAGRAM

Conceptual Design



Explain the term 'Degree'

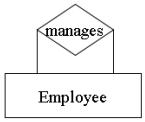
- **Number of participating entity types.**

Degree of Relationship Type

- **Unary Relationship**
  - **A relationship between the instances of a single entity type**

    e.g. *Person is married to a Person*

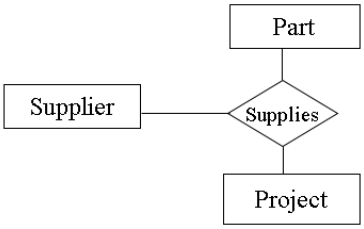
    Employee manages Employees



- **Binary Relationship**
  - **A relationship between the instances of two entity types**

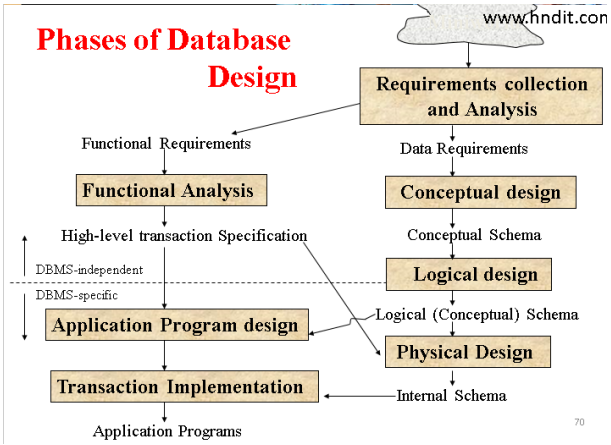
    e.g. *An employee works for a department*

- **Ternary Relationship**
- **A simultaneous relationship among the instances of three entity types**
  - **Supplier *s* supplies part *p* to project *j***



Cardinality in DBMS

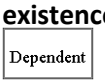
- **Defines the maximum number of relationship instances in which an entity can participate.**
- **The possible cardinality ratios for binary relationship types are**
  - **1:1**
  - **1:N**
  - **M:N**



Entity Types

- **Strong (Regular) Entity**
  - **An entity that exists independently of other entity types**

- **Weak Entity**
  - **An entity types whose existence depends on some other entity**



- **Identifying Owner**
  - **The entity type on which the weak entity type depends**
  - *e.g. Employee is the Owner of Dependent*
- **Identifying Relationship**
  - **A relationship between a weak entity type and its owner**

Attributes

- **Multi-valued Attribute**
  - **An attribute that may take on more than one value for a given entity instance**
  - *e.g. Employee Skills, Qualifications*
- **Composite Attribute**
  - **An attribute that can be broken down into component parts**
  - *e.g. Address (Street, City, State, Postal Code)*
  - *Name (First Name, Middle Initials, Last Name)*

Key Attribute (Identifier)

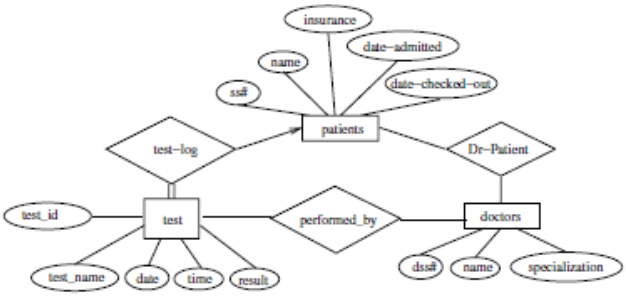
- ***Identifier***
  - **An attribute (or combination of attributes) that uniquely identifiers individual instances of an entity type**
  - *e.g. Emp No*
- ***Composite Identifier***
  - **An identifier that consists of a composite attribute**
  - *e.g. Flight Id (Flight No, Date)*

Associative entities

- *are used when you need a relationship to be involved in a relationship.*

ERD Example 1

Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted.



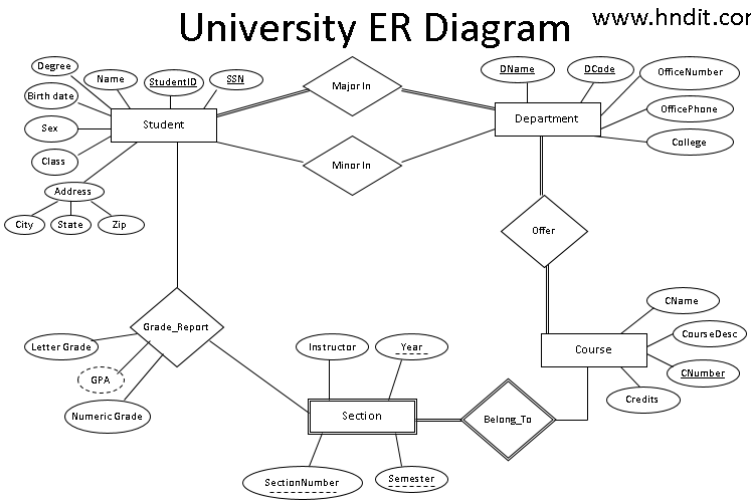
Hospital tables:

- **patients (patient-id, name, insurance, date-admitted, date-checked-out)**
- **doctors (doctor-id, name, specialization)**
- **test (testid, testname, date, time, result)**
- **doctor-patient (patient-id, doctor-id)**
- **test-log (testid, patient-id) performed-by (testid, doctor-id)**

ERD Example 2

Consider the following set of requirements for a University database. Design an ER diagram for this application:

- The university keeps track of each student's name, student number, social security number, current address and phone number, permanent address and phone number, birthdate, sex, class (freshman, graduate), major department, minor department (if any), degree program (B.A., B.S., ... Ph.D.). Some user applications need to refer to the city, state, and zip code of the student's permanent address and to the student's last name. Both social security number and student number are unique for each student. All students will have at least a major department.
- Each department is described by a name, department code, office number, office phone, and college. Both the name and code have unique values for each department.
- Each course has a course name, description, course number, number of credits, level and offering department. The course number is unique for each course.
- Each section has an instructor, semester, year, course, and section number. The section number distinguishes sections of the same course that are taught during the same semester/year; its value is an integer (1, 2, 3, ... up to the number of sections taught during each semester).
- A grade report must be generated for each student that lists the section, letter grade, and numeric grade (0,1,2,3, or 4) for each student and calculates his or her average GPA.



Method of describing the process

- **Structured English**
- **Decision tree**
- **Decision Table**

Structured English

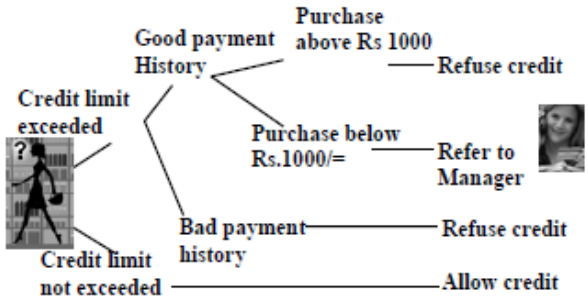
IF credit limit exceeded
THEN
  IF customer has bad payment history
  THEN refuse credit
  ELSE
  IF purchase above Rs. 1000/=
  THEN refuse credit
  ELSE refer to manager
ELSE allow credit

- **Structured English used to**
  - **Explain the <u>conditions</u> which occurs in a process.**
  - **Identify the <u>decisions</u> which makes these conditions occur.**
  - **Find <u>alternative</u> actions to be taken**

- **The process is defined by using three types of statements(Developing Structure Statements)**
  - **Sequence structure**
  - **Decision structure**
  - **Iteration structure**

Decision Tree

- **A decision tree can help you examine all possible options when faced with a hard choice or decision.**



Decision table

- A decision table lists causes and effects in a matrix. Each column represents a unique combination.

  Cause = condition

  effect = action = expected results

- If all causes are simply Y/N values:

  Number of combinations = $2^{\text{number of causes}}$

		Combinations							
Causes	**Values**	1	2	3	4	5	6	7	8
Cause 1	*Y, N*	Y	Y	Y	Y	N	N	N	N
Cause 2	*Y, N*	Y	Y	N	N	Y	Y	N	N
Cause 3	*Y, N*	Y	N	Y	N	Y	N	Y	N
Effects									
Effect 1		X			X				X
Effect 2			X				X		X

Steps to Create a decision table

- **List all causes (conditions) in the decision table**
- **Calculate the number of possible combinations**
- **Fill columns with all possible combinations**
- **Reduce test combinations**
- **Check covered combinations**

Using Decision Table

Condition	Credit limit exceeded	Y	Y	Y	Y	N	N	N	N
	Good payment history	Y	Y	N	N	Y	Y	N	N
	Purchase above Rs.1000/=	Y	N	Y	N	Y	N	Y	N
Action	Allow Credit					X	X	X	X
	Refuse	X		X	X				
	Refer Manager		X						

Y-TRUE N-NOT TRUE X-TAKE ACTION

System design Approaches

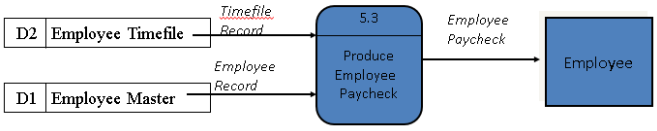
- **Modern structured design**
- **Information engineering**
- **Prototyping**
- **JAD**
- **RAD**
- **Object-oriented**

(S10)System Design-II

Data Dictionary

- **A document describing a database or collection of databases ·**
  (contains metadata i.e data about the database.)

Data Dictionary



Date Flow Diagram

Employee Record = Employee Number + Personal information+ Wage information + Current pay information+	Timefile Record = Employee Number + Employee Name + Hours worked
	Employee Paycheck = Employee Number + Employee Name + Address+ Current Pay Amounts

Data Structure

Synchronization

- **Is the process of maintaining consistency between the different types of models**

(S11)System Testing

Types of Tests

- **Black-Box Testing**
- **White-Box Testing**
- **Gray-Box Testing**

Black-box testing

- **takes an external perspective of the test object to derive test cases.**
- **Typical black box test design techniques include:**
  - **Decision table testing**
  - **Pair wise testing**
  - **State transition tables**
  - **Use case testing**
  - **Cross-functional testing**

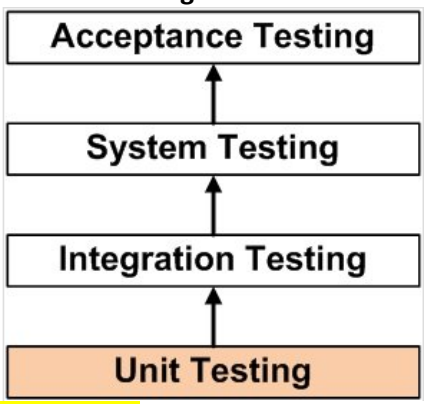
White-box testing

- **uses an internal perspective of the system to design test cases based on internal structure.**
- **Typical white box test design techniques include:**
  - **Control flow testing**
  - **Data flow testing**
  - **Branch testing**

Gray Box Testing

- **is a software testing method which is a combination of Black Box Testing method and White Box Testing method.**

Levels of testing



Unit testing

- **unit testing is a software verification and validation method in which a programmer tests if individual units of source code are fit for use.**
- **performed by using the White Box Testing method**

verification and validation

Criteria	Verification	Validation
Definition	The process of evaluating work-products (not the actual final product) of a development phase to determine whether they meet the specified requirements for that phase.	The process of evaluating software during or at the end of the development process to determine whether it satisfies specified business requirements.
Question	Are we building the product *right*?	Are we building the *right* product?
Evaluation Items	Plans, Requirement Specs, Design Specs, Code, Test Cases	The actual product/software.
Activities	•Reviews •Walkthroughs •Inspections	•Testing

Benefits of unit testing

- Facilitates change
- **allows the programmer to refractor code at a later date**
- Simplifies integration
- **reduce uncertainty in the units themselves**
- Documentation
  - provides a sort of living documentation of the system

Limitations of unit testing

- Software testing is a combinatorial problem
- Testing cannot be expected to catch every error in the program
  (not catch integration errors or broader system-level errors)

Integration testing

- phase in software testing in which individual software modules are combined and tested as a group.
- The purpose verify functional, performance, and reliability requirements placed on major design items.

METHOD

- **Any of Black Box Testing, White Box Testing, and Gray Box Testing methods can be used. Normally, the method depends on your definition of 'unit'.**

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Integration testing cont..(different types)

- top-down
- *to integrated testing where the top integrated modules*
- bottom-up
- *to integrated testing where the lowest level components*

System Testing

- **level of the software testing process where a complete, integrated system/software is tested.**

METHOD

- **Black Box Testing method is used.**

Who performs it

- **independent Testers perform System Testing.**

User acceptance testing

- **is the final stage of any software development or change request lifecycle before go-live.**

Test automation

- Automated testing tools are sometimes referred to as Computer Aided Software Testing (CAST) tools.
- use of software to
  - control the execution of tests
  - the comparison of actual outcomes to predicted outcomes
  - the setting up of test preconditions
  - test reporting functions.
- two general approaches
  - Code-driven testing
  - Graphical user interface testing.

Testing Strategy

- cover the following areas.
  - Strategy approach-
    A testing strategy should be formulated
  - Test plan –
    A test plan should be developed
  - Test design –
    The logic and reasoning behind the design of the tests should be explained.
  - Performing tests –
    Detailed procedures should be provided for all tests.
  - Documentation –
    It must be clear how the results of tests are to be documented.
  - Re-testing –
    The re-test procedure should be explained.

The limitations of software testing

- Poor testing process
- Inadequate time
- Future requirements not anticipated
- Inadequate test data
- Software changes inadequately test

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(S12)System Implementation

Includes

- Source code
- Database
- User documentation
- Testing
- Staff training
- File conversion

Source code may be in one of the following categories

- Standard off- the –shelf package
- Bespoke package
- Amended standard package
- standard package plus additions

Standard off- the –shelf package

- **The organization purchases and installs a ready-made solution.**
- **Advantages**
- *Available immediately*
- *Cheaper*
- *High quality*
- *Can be update*
- *Relatively free of bugs*
- *Easy to handle*

- **DisAdvantages**
- It may be not well suited
- Competitors may well use the same package.

The factors to consider when choosing *an off-the –shelf* application package

- User requirements -
- Processing times
- Documentation
- Compatibility
- Controls
- User-interface
- Modification
- cost
- Support maintenance and update

Bespoke package

- Programmers write an application to meet the specific needs of the organization.

Advantage

- The software should meet the organization's specific needed.
- Competitive advantage
- It can make modification for future needs

Disadvantages:

  - It make risk
  - Greater chance of bugs
  - Waste time and cost

Amended standard package

- **This may require access to the source code.(for some customization)**
- **Development time should be much quicker.**
- **User can get good knowledge about the software**

standard package plus additions

- **The purchased standard package is not amended itself,**
- **but additional software that integrates with the standard package is developed.**
- **This also may require access to the source code.**

User documentation

- Written or visual information about an application system, how it works, and how to use it.

Technical manual

- should include the following;
  - Hardware technical specification
  - System objectives
  - Flowcharts or Data flow diagrams
  - Entity models and life histories
  - Individual program specifications
  - Data dictionary
  - Contact details for the original developers
  - System overview
  - System specifications including performance details

User manual

- The manual provides full documentation of the operational procedures necessary for the 'hands-on' running of the system;
- Systems set-up procedures
- Security procedures
- Reconstruction control procedures
- System messages
- Samples

Staff training

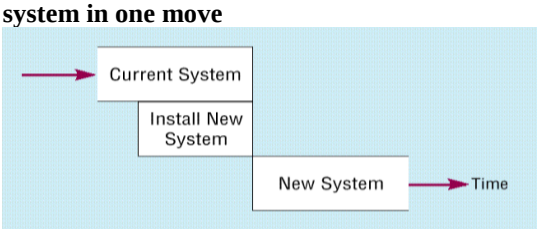
- Senior management training
  - Executive Support System and Decision Support System
  - Awareness of information technology, project mgt. skills
- Middle managers training
  - Computing skill
  - Management information system
  - Office type software
- Operational staff
  - Training should focus on specific tasks

File conversion

- This may be according to one of four approaches
  1. Direct changeover
  2. Parallel running
  3. Pilot operation
  4. Phased changeover

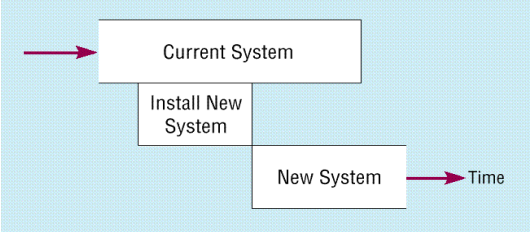
1. Direct changeover

- **The old system is completely replaced by the new system in one move**



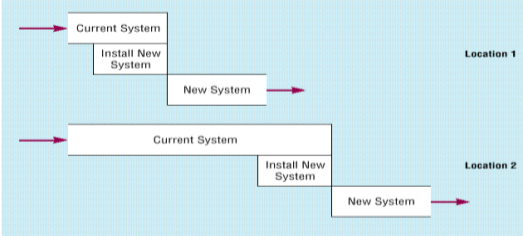
2. Parallel running

- **The old and new systems are run in parallel for a period of time.**



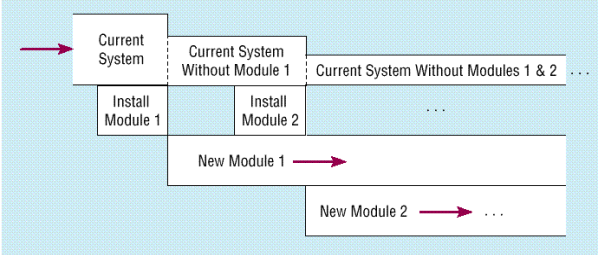
3.pilot operation

- **involves selecting parts of an organization to operate running the new system in parallel with the existing system.**



4.Phased changeover

- **Selecting a complete section of the system for a direct changeover.Introduce the rest phase by phase.**



Method	Advantages	Disadvantages
Direct changeover	Quick Minimal cost Minimizes workload	Risky Could disrupt operations If fails, will be costly
Parallel running	Safe, built-in safety Provides way of verifying results of new system	Costly-two systems need to be operated Time –consuming Additional workload

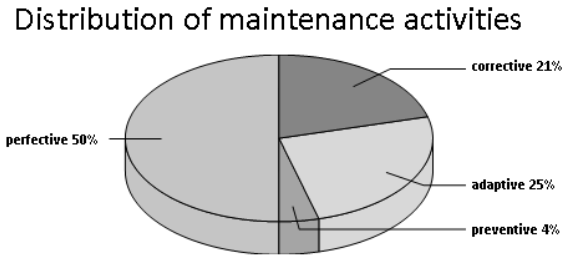
Method	Advantages	Disadvantages
Pilot operation	Less risky than direct changeover Less costly than complete parallel running	Can take a long time to achieve Total changeover Not as safe as complete parallel running
Phased changeover	Less risky than a single direct changeover Any problems should be in one area-other operations unaffected	Can take a long time to achieve total changeover Interfaces between parts of the system may make this impractical

(S13): System Maintenance

- **performing hardware and software installation, diagnostic and correction and similar activities that are performed in the usual .**

Kinds of maintenance activities

- ***corrective maintenance*: correcting errors**
- ***adaptive maintenance*: adapting to changes in the environment (both hardware and software)**
- ***perfective maintenance*: adapting to changing user requirements**
- ***preventive maintenance*: increasing the system's maintainability**



Key to maintenance is in development

- **Higher quality ⇒ less maintenance (corrective maintenance)**
- **Anticipating changes ⇒ less maintenance (adaptive and perfective maintenance)**
- **Better tuning to user needs ⇒ less maintenance (perfective maintenance)**
- **Less code ⇒ less maintenance**

Shift in type of maintenance over time

- **Introductory stage: emphasis on user support**
- **Growth stage: emphasis on correcting faults**
- **Maturity: emphasis on enhancements**
- **Decline: emphasis on technology changes**

Major causes of maintenance problems

- **Unstructured code**
- **Insufficient domain knowledge**
- **Insufficient documentation**